Pediatric Reference Ranges

Age Group: Under 10 Years

Reference ranges for children under 10 reflect active growth phases. Differences between biological sexes are generally minimal before the onset of puberty, so many baseline targets are shared.

Biomarker Reference Table (< 10 Years)

Parameter	Male Range	Female Range	Units
Hemoglobin (Hb)	11.5 – 14.5	11.5 – 14.5	g/dL
RBC_Count	4.0 – 5.2	4.0 – 5.2	million/mcL
WBC_Count	5,000 – 14,500	5,000 – 14,500	cells/mcL
Platelet_Count	150,000 – 450,000	150,000 – 450,000	platelets/mcL
Fasting_Blood_Sugar	70 – 100	70 – 100	mg/dL
Total_Cholesterol	< 170	< 170	mg/dL
HDL (Good)	> 45	> 45	mg/dL
LDL (Bad)	< 110	< 110	mg/dL
Triglycerides	< 90	< 90	mg/dL

Health_Flag Note: Pediatric WBC counts are naturally higher than adults due to a developing immune system. High flags here may just indicate normal childhood immune responses.

Adolescent Reference Ranges

Age Group: 10 to 19 Years (Under 20)

During adolescence, sex hormones begin to drive divergent physiological changes. Hemoglobin and red blood cell counts begin to separate significantly between males and females.

Biomarker Reference Table (< 20 Years)

Parameter	Male Range	Female Range	Units
Hemoglobin (Hb)	13.0 – 16.0	12.0 – 15.0	g/dL
RBC_Count	4.5 – 5.3	4.1 – 5.1	million/mcL
WBC_Count	4,500 – 11,000	4,500 – 11,000	cells/mcL
Platelet_Count	150,000 – 450,000	150,000 – 450,000	platelets/mcL
Fasting_Blood_Sugar	70 – 100	70 – 100	mg/dL
Total_Cholesterol	< 170	< 170	mg/dL
HDL (Good)	> 45	> 45	mg/dL
LDL (Bad)	< 110	< 110	mg/dL
Triglycerides	< 130	< 130	mg/dL

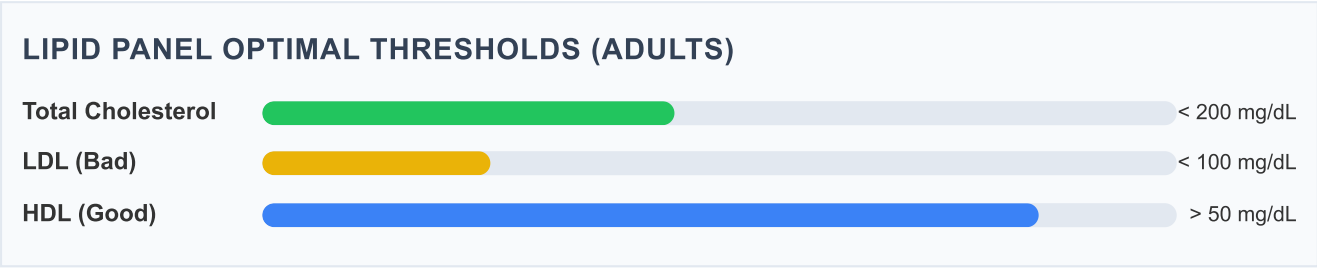
Health_Flag Note: Menstruation in adolescent females can lower Hb and RBC counts. Clinical health flags for anemia are calibrated to these newly divergent adolescent baselines.

Young Adult Reference Ranges

Age Group: 20 to 29 Years (Under 30)

Biomarker Reference Table (< 30 Years)

Parameter	Male Range	Female Range	Units
Hemoglobin (Hb)	13.8 – 17.2	12.1 – 15.1	g/dL
RBC_Count	4.7 – 6.1	4.2 – 5.4	million/mcL
WBC_Count	4,500 – 11,000	4,500 – 11,000	cells/mcL
Platelet_Count	150,000 – 450,000	150,000 – 450,000	platelets/mcL
Fasting_Blood_Sugar	70 – 99	70 – 99	mg/dL
Total_Cholesterol	< 200	< 200	mg/dL
HDL (Good)	> 40	> 50	mg/dL
LDL (Bad)	< 100	< 100	mg/dL
Triglycerides	< 150	< 150	mg/dL



Adult Reference Ranges

Age Group: 30 to 39 Years (Under 40)

Metabolic baselines stabilize in this decade. Preventative monitoring focuses on the early detection of lipid imbalances and fasting blood sugar creep, which present diverging risk profiles between sexes.

Biomarker Reference Table (< 40 Years)

Parameter	Male Range	Female Range	Units
Hemoglobin (Hb)	13.8 – 17.2	12.1 – 15.1	g/dL
RBC_Count	4.7 – 6.1	4.2 – 5.4	million/mcL
WBC_Count	4,500 – 11,000	4,500 – 11,000	cells/mcL
Platelet_Count	150,000 – 450,000	150,000 – 450,000	platelets/mcL
Fasting_Blood_Sugar	70 – 99	70 – 99	mg/dL
Total_Cholesterol	< 200	< 200	mg/dL
HDL (Good)	> 40	> 50	mg/dL
LDL (Bad)	< 100	< 100	mg/dL
Triglycerides	< 150	< 150	mg/dL

Metabolic Flags

Fasting sugars consistently >99 trigger pre-diabetic flags. Monitoring HbA1c may be recommended.

Cardiovascular Flags

Triglycerides >150 and LDL >100 often result in preventative lipid-management flags in males.

Mid-Adulthood Ranges

Age Group: 40 to 49 Years (Under 50)

As metabolic rates slow, strict adherence to optimal lipid and glucose ranges is emphasized. The standard healthy intervals remain consistent with younger adults, but flag thresholds are monitored more strictly.

Biomarker Reference Table (< 50 Years)

Parameter	Male Range	Female Range	Units
Hemoglobin (Hb)	13.8 – 17.2	12.1 – 15.1	g/dL
RBC_Count	4.7 – 6.1	4.2 – 5.4	million/mcL
WBC_Count	4,500 – 11,000	4,500 – 11,000	cells/mcL
Platelet_Count	150,000 – 450,000	150,000 – 450,000	platelets/mcL
Fasting_Blood_Sugar	70 – 99	70 – 99	mg/dL
Total_Cholesterol	< 200	< 200	mg/dL
HDL (Good)	> 40	> 50	mg/dL
LDL (Bad)	< 100	< 100	mg/dL
Triglycerides	< 150	< 150	mg/dL

Late Adulthood Ranges

Age Group: 50 to 59 Years (Under 60)

This decade includes menopause for many females, which can slightly elevate baseline hemoglobin levels as menstrual blood loss ceases. Males may experience gradual declines in testosterone affecting RBC synthesis.

Biomarker Reference Table (< 60 Years)

Parameter	Male Range	Female Range	Units
Hemoglobin (Hb)	13.5 – 17.0	12.5 – 15.5	g/dL
RBC_Count	4.5 – 5.9	4.2 – 5.4	million/mcL
WBC_Count	4,500 – 11,000	4,500 – 11,000	cells/mcL
Platelet_Count	150,000 – 450,000	150,000 – 450,000	platelets/mcL
Fasting_Blood_Sugar	70 – 99	70 – 99	mg/dL
Total_Cholesterol	< 200	< 200	mg/dL
HDL (Good)	> 40	> 50	mg/dL
LDL (Bad)	< 100	< 100	mg/dL
Triglycerides	< 150	< 150	mg/dL

Health_Flag Note: Post-menopausal females lose the protective cardiovascular effects of estrogen, making HDL and LDL flags a critical monitoring focus in this age bracket.

Geriatric Reference Ranges

Age Group: 60+ Years

In senior populations, bone marrow production naturally slows down, leading to mildly lower normal ranges for Hemoglobin and Red Blood Cells compared to younger adults.

Biomarker Reference Table (60+ Years)

Parameter	Male Range	Female Range	Units
Hemoglobin (Hb)	12.4 – 14.9	11.7 – 13.8	g/dL
RBC_Count	4.2 – 5.6	3.8 – 5.2	million/mcL
WBC_Count	4,500 – 10,000	4,500 – 10,000	cells/mcL
Platelet_Count	150,000 – 400,000	150,000 – 400,000	platelets/mcL
Fasting_Blood_Sugar	70 – 105	70 – 105	mg/dL
Total_Cholesterol	< 200	< 200	mg/dL
HDL (Good)	> 40	> 50	mg/dL
LDL (Bad)	< 100	< 100	mg/dL
Triglycerides	< 150	< 150	mg/dL

Hematology Note

Anemia flags are common but should not be dismissed as solely age-related without clinical investigation.

Metabolic Note

Fasting glucose normal ranges may be slightly relaxed up to 105 mg/dL depending on physician discretion.